

Program : Diploma in Chemical Engineering	
Course Code : 3078	Course Title: Engineering chemistry Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To impart knowledge on basic engineering chemistry experiments through volumetric analysis.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basic knowledge in volumetric analysis.		Applied chemistry Lab. I	I
The reactions of acid and basic radicals.		Applied chemistry Lab. II	II

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Practice Acidimetric and Alkalimetric analysis.	6	Applying
CO2	Practice Permanganometry	15	Applying
CO3	Practice Dichrometry.	9	Applying
CO4	Practice Iodometry, iodimetry and Argentometry	15	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Practice Acidimetric and Alkalimetric analysis.		
M1.01	Estimate the quantity of sulphuric acid present in the given sample of solution using oxalic acid crystals and deci normal solution of NaOH.	3	Applying
M1.02	Estimate the amount of sodium hydroxide present in the given sample of solutions using deci normal solution of HCl and AR sodium carbonate crystals.	3	Applying
CO2	Practice Permanganometry		
M2.01	Estimate the amount of sodium oxalate in the given sample of solution using deci normal solution of potassium permanganate and oxalic acid crystals.	3	Applying
M2.02	Estimate the amount of oxalic acid in the given sample of solution using AR Mohr's salt crystals and deci normal solution of potassium permanganate.	3	Applying
M2.03	Estimate the amount of Mohr's salt in the given sample of solution using pure oxalic acid crystals and deci normal solution of potassium permanganate.	3	Applying
M2.04	Estimate the amount of ferrous ion in the given sample of solution using AR Mohr's salt crystals and deci normal solution of potassium permanganate.	3	Applying
	Lab Exam – I	3	

CO3	Practice Dichrometry.		
M3.01	Estimate the amount of ferrous ion in the given sample of solution using AR Mohr's salt and deci normal solution of potassium dichromate using internal indicator.	3	Applying
M3.02	Estimate the amount of ferrous ion in the given sample of solution using potassium dichromate crystals using external indicator.	3	Applying
M3.03	Open ended experiments	3	Applying
CO4	Practice Iodometry, iodimetry and Argentometry		
M4.01	Estimate the amount of potassium dichromate in the whole of the given solution, being supplied with approximately N/20 sodium thiosulphate solution and electrolytic copper.	3	Applying
M4.02	Estimate the amount of copper in the whole of the given solution supplied with pure potassium dichromate crystals and approximately N/20 solution of sodium thiosulphate.	3	Applying
M4.03	Estimation of bleaching powder.	3	Applying
M4.04	Estimate the amount of chloride in the whole of the given solution of potassium chloride, supplied with approx. N/20 silver nitrate solution and AR sodium chloride crystals.	3	Applying
	Lab Exam – II		

Text / Reference

T/R	Book Title/Author
T1	Practical Chemistry for B. Sc Main ; A. O. Thomas.
R1	Applied Chemistry Laboratory Practices, Vol. I and Vol. II; Dr. G. H. Hugar & Prof A. N. Pathak

Online Resources

Sl.No	Website Link
1	https://www.vlab.co.in/broad-area-chemical-sciences